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17. Disinfectants An experiment was conducted to study the use of 95% ethanol versus 20% bleach as a disinfectant in removing contamination when culturing plant tissues. The experiment was repeated 15 times with each disinfectant, using cuttings of eggplant tissue.⁹ The observation reported was the number of uncontaminated eggplant cuttings after a 4-week storage.

Disinfectant	95% Ethanol	20% Bleach
Mean	3.73	4.80
Variance	2.78095	.17143
n	15	15
Pooled variance 1.47619		

- Are you willing to assume that the underlying variances are equal?
- Using the information from part a, is there evidence of a significant difference in the mean numbers of uncontaminated eggplants for the two disinfectants tested?

20. Dissolved O₂ Content A pollution control inspector suspected that a river community was releasing

$$a. \frac{s_1}{s_2} = \frac{2.78095}{0.17143} > 3 \Rightarrow \text{assume not equal}$$

$$a. \quad k = 1, \quad \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$